MIL-E-1/1022D
22 February 1977
SUPERSEDING
MIL-E-1/1022C
16 June 1971
(See note 9)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6384

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, beam power, pulsed operation

Outline --- See figure 3

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	3	5	6	7	8
Element	---	k, g3	a	g2	h	gl	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ecl	Ec2	Ehk	Rgl	Ik	Icl	Pp	Pg2	epy	ib	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Meg	mAdc	mAdc	W	W	v	a	°C	ft
Maximum:	6.9	750	0, -200	325	450	0.1	125	+5.0	30	3.5	1,500	1.0 (See figure 1)	300	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-22.5	250	0	---	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Not required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	1.14	1.26	A
1336	Heater-cathode leakage	-	Ehk = 450 Vdc	0.4	II	Ihk	---	10	µAdc
---	High voltage	2	Ec1 = -150 Vdc; Ec2 = 300 Vdc; Ebb = 1,250 Vdc Ic1 = 1.0 µAdc (max); Rk = 5,000 ohms	0.65	II	e	1,150	---	v
1266	Total grid current (1)	-		0.65	II	Ic1	---	-0.2	µAdc
1256	Electrode current (1) (anode)	3		0.65	II	Ib	66	88	mAdc
1256	Electrode current (2) (anode)	-	Ec1 = -60 Vdc	0.65	II	Ib	---	0.5	mAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	0.5	6.5	mAdc
1306	Transconductance (1)	3		0.65	II	Sm	4,800	6,000	µmhos
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-	Ef = 6.3 V g1 to all a to all	4.0	S3	R R	300 500	--- ---	Meg Meg
1306	Transconductance (2)	-	Ef = 5.7 V	---	---	Δ Sm Ef	---	5	%
1266	Total grid current (2)	4	Ef = 7.0 V	---	---	Ic1	---	-0.3	µAdc
1256	Electrode current (3) (anode)	-	Ec1 = -40 Vdc	---	---	Ib	5	---	mAdc
1246	Audio frequency noise	-	Ec1 = -35 Vac; Rp = 2,000 ohms; Esig = 120 mVac	---	---	EB	---	17	vu
1266	Primary grid emission (screen)	5	Eg2 = 150 Vac; Eb = 0	---	---	Ic2	---	-750	µAdc
1331	Direct-interelectrode capacitance	-	No shield	---	---	Cgla Cin Cout	0.5 8.5 5.0	3.0 13.5 9.0	pF pF pF
1311	Anode resistance	-		---	---	rp	0.015	---	Meg
1316	Amplification factor	-	Eb = Ec2 = 250 Vdc; Ec1 = -22.5 Vdc	---	---	Mu	5.5	6.5	---
---	Pulsed operation	6	Ebb = 750 Vdc; Ecc1 = -80 Vdc; Ecc2 = 325 Vdc; ec1 = +100 v	---	---	ib	1.1	---	a
1041	Shock	-	450 G; Ehk = +450 Vdc; Ec1 = 0; Rk = 300 ohms; Rg1 = 0.1 Meg	---	---	---	---	---	---
1031	Vibration fatigue	7	F = 50 to 500 to 50 Hz; time of sweep 15 min; 5 G	6.5	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
---	Shock and vibration-fatigue test end points:								
1336	Heater-cathode leakage	-	Ehk = 450 Vdc	---	---	Ihk	---	15	µAdc
1266	Total grid current (1)	-		---	---	Ic1	---	-0.5	µAdc
1306	Transconductance (1)	-		---	---	Sm	4,500	---	µmhos
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Intermittent life	8	Group A; Ec1 = -18 Vdc; Ehk = 300 Vac; Rg1 = 47,000 ohms; TE = 300°C (min)	---	---	---	---	---	---
---	Intermittent life-test end points (1,000 hours):								
---	Inoperatives	-		---	---	---	---	---	---
1336	Heater-cathode leakage	-	Ehk = 450 Vdc	---	---	Ihk	---	25	µAdc
1266	Total grid current (1)	-		---	---	Ic1	---	-0.5	µAdc
1306	Transconductance (1)	-		---	---	Sm	4,500	6,000	µmhos
1306	Transconductance (1), average change	-		---	---	ΔSm_t	---	10	%
1211	Insulation of electrodes	-	g1 to all a to all	---	---	R R	150 250	--- ---	Meg Meg
---	Pulse life	-	Group B; Ecc1 = -80 Vdc; Ecc2 = 325 Vdc; Ebb = 750 Vdc; adjust ec1 to give ib = 1.0 a; tp = 10 µs; prr = 1,000	---	---	---	---	---	---
---	Pulse life-test end points (500 hours):								
---	Inoperatives	-		---	---	---	---	---	---
1266	Total grid current (1)	-	t = 120 (max)	---	---	Ic1	---	-0.5	µAdc
---	Pulsed operation	-		---	---	ib	0.9	---	a

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test is to be made before the transconductance (1) and transconductance (2) tests. With the specified anode supply voltage applied to the tube through 5,000 ohms and with a 60 Hz sinusoidal drive voltage applied to the control grid, measure the peak-to-peak voltage across the load.
3. This test to be performed at the conclusion of the holding period.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes operating at the conditions specified below. The 3-minute test is not permitted. Test at preheat conditions within 3 seconds after preheating. Total grid current (2) shall be the last test performed on the sample selected for the total grid current (2) test.

Ef	Ec1	Ec2	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Ohms	Meg
7.0	0	250	250	250	0.1

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NOTES: -Continued

5. Operate the tube for 5 minutes at test conditions prior to the primary grid emission (screen) test. With the specified 60 Hz sinusoidal voltage applied to the screen, adjust Ec1 to provide Ic2 = 21 mAdc on the positive half cycles. The primary emission shall be measured on the negative half cycles and shall not exceed the limit specified.

6. See circuit diagram figure 2. Use rectangular wave modulation on the grid. Repetition rate = 1,000 pps. Pulse width = 10 microseconds. Apply anode and screen voltages simultaneously. The ib shall be measured after a minimum operating time of 5 minutes.

7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

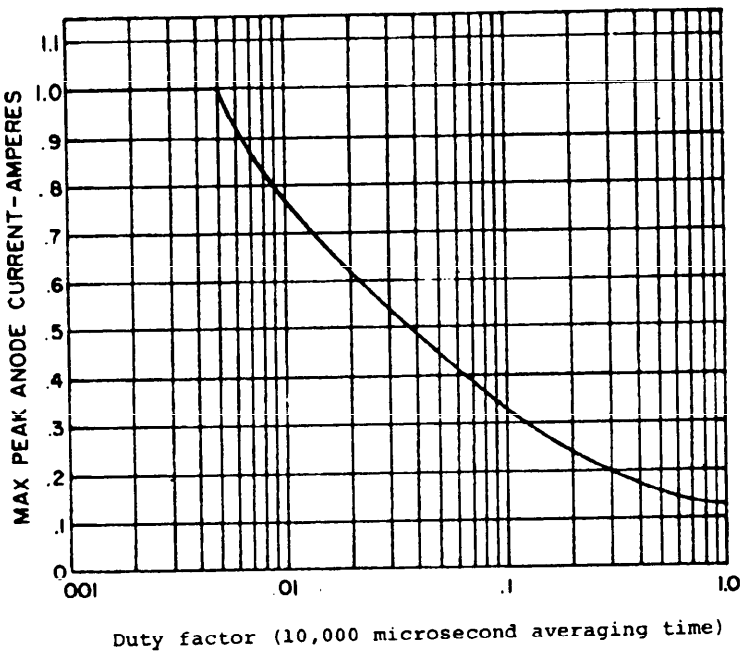
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG
Air Force - 11, 17

Preparing activity:
Navy - EC

Agent:
DLA - ES

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Duty factor: For the 6384 is defined as the ratio of 'ON' time in microseconds to 10,000 microseconds.

'ON' time: Is defined as the sum of the duration of all individual pulses which occur during any 10,000-microsecond interval.

FIGURE 1. Pulse rating chart.

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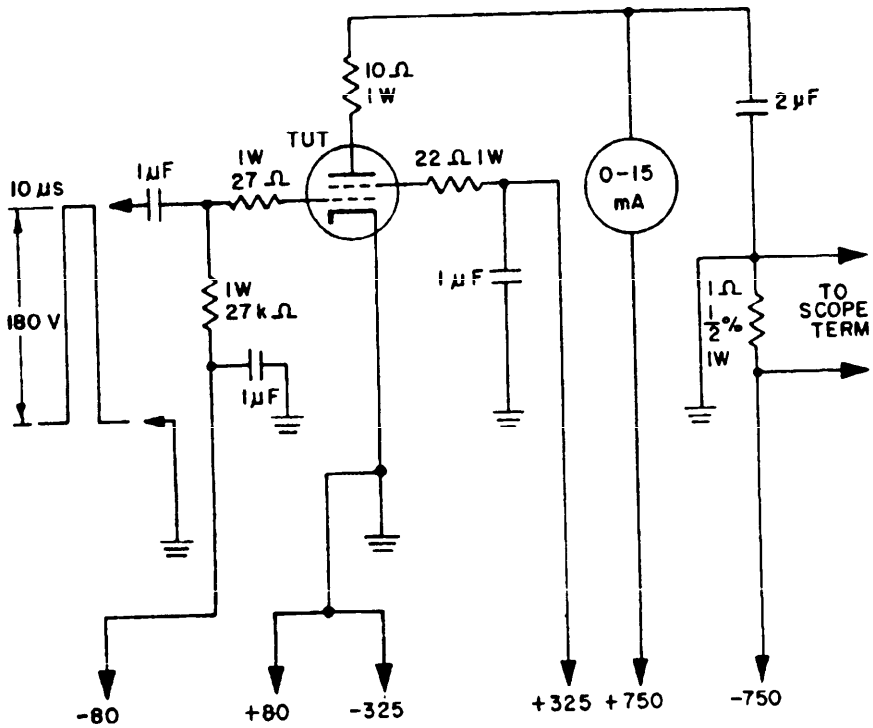
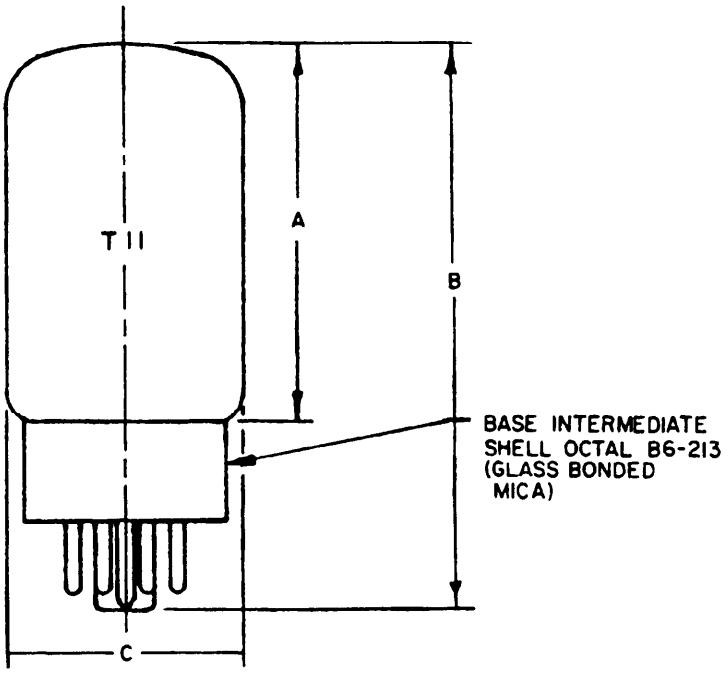


FIGURE 2. Pulsed operation circuit diagram.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		2.94(74.68)
B		3.47(88.14)
C		1.44(36.58)

FIGURE 3. Outline drawing of electron tube type 6384.

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